Benchmark Review Report

1. Problem Framing

Detection task: Instance segmentation of rip currents. The paper specifically moves beyond bounding box detection — which only produces a rough rectangular estimate — to full polygon-level segmentation that captures the precise shape, position and extent of each rip current.

Why segmentation over bounding boxes: Rip currents are amorphous by nature — they have no fixed shape. Bounding boxes often include large areas of irrelevant background water or exclude parts of the rip current channel. Polygonal instance segmentation solves both problems by tracing the actual boundary.

Intended user context: Beach safety systems, lifeguard monitoring, drone-based surveillance, and eventually mobile device alerts for the general public. The paper explicitly targets real-time detection for practical deployment.

2. Dataset Details

Field	**Details**
Dataset name	Dumitriu et al. 2023 Rip Current Benchmark
Public link	https://github.com/Irikos/rip_currents
Training images	2,466 images with polygonal annotations + 1,307 images without rip currents
Test videos	17 drone videos — 24,295 total frames at 30 FPS (~29 minutes of footage)
Video resolutions	2 videos at 3840×2160 (4K); remainder at 1920×1080
Location	Black Sea beach, Eforie Nord, Romania — recorded across 2021, 2022 and 2023

Viewpoint	Aerial drone footage — mostly hovering fixed standpoint; some moving shots at different elevations
Conditions	Various dates and times; f2.8 aperture, ISO 100 for 1080p videos
Annotation type	Polygonal masks for instance segmentation + axis-aligned bounding boxes for object detection
Annotation tool	Roboflow polygon tool — verified by a co-author who is a rip current domain expert
License	CC BY-NC 4.0 — non-commercial use only

Note on annotation difficulty: Because rip currents are amorphous, annotations required informed expert assumptions in cases where the shape was not fully visible. The dataset also includes images from Google Earth (satellite view) and ground-level beachfront photos, creating diversity in viewpoint and scale.

3. Method / Model

Model family: YOLOv8 (You Only Look Once, version 8) — an anchor-free, single-stage object detection and segmentation model by Ultralytics. Five sizes were evaluated: nano (n), small (s), medium (m), large (l), and extra-large (x).

Why YOLOv8 was chosen over alternatives: The authors considered state-of-the-art models including EVA, BEiT-3, Mask DINO, and Mask R-CNN, but selected YOLOv8 for three reasons: (1) proven real-time detection performance essential for public safety, (2) large user community and accessible implementation, and (3) easy deployment across multiple sizes for hardware flexibility.

Special components: None — YOLOv8 was used out-of-the-box without modifications. This was a deliberate choice to establish a clean, reproducible baseline.

Temporal handling: The model was trained on static images only. No temporal aggregation or video-aware processing was used. The authors acknowledge this as a limitation and suggest future work should incorporate temporal information from video frames.

Explainability: No heatmaps, Grad-CAM or overlay visualisations were included. The paper focuses purely on detection performance metrics.

Training environment: NVIDIA GeForce RTX 3090 (24GB VRAM), Python 3.10.4, PyTorch 1.12.1, CUDA 11.7. Trained at 640×640 pixel resolution for up to 300 epochs with early stopping patience of 50 epochs.

4. Evaluation Setup

Train / Validation Split

A 10-fold stratified cross-validation was applied to the training image dataset (2,466 rip + 1,307 non-rip images), using a 90% train / 10% validation split per fold. Stratified sampling ensured consistent class distribution across all folds. The test set was kept entirely separate — comprising the 17 drone videos plus 5 beachfront no-rip videos from De Silva et al.

Leakage risk: Low for the image/video split — training was on static images and testing was on entirely separate drone videos. However, some training images originate from the same beach location as the test videos, which may allow the model to partially overfit to specific environmental conditions at Eforie Nord.

Metrics Reported

- mAP50 — primary metric. Mean Average Precision at IoU threshold of 0.50
- mAP50:95 — stricter metric averaging mAP across IoU thresholds from 0.50 to 0.95 in 0.05 increments
- Precision — true positives / (true positives + false positives)
- Recall — true positives / (true positives + false negatives)
- FPS — frames per second, used to assess real-time viability
- Standard deviation of mAP50 across 10 folds — measures training stability

Test results reported as both macro average (mean across all videos regardless of length) and micro average (weighted by number of frames per video).

Baseline Comparisons

All five YOLOv8 variants (n, s, m, l, x) were compared against each other. No external baseline models (e.g. Faster R-CNN, YOLO-Rip) were directly benchmarked on this dataset, as this paper introduces the dataset itself. Prior works used different datasets making direct comparison impossible.

5. Performance + Limitations

Headline Results

Model	FPS	Precision	Recall	mAP50 (val)	mAP50 (test macro)	mAP50:95
YOLOv8n	73.33	88.27%	84.67%	88.94%	81.21%	45.75%
YOLOv8s	33.74	87.62%	84.67%	89.25%	68.11%	46.66%
YOLOv8m	15.54	88.04%	85.04%	89.92%	76.27%	47.48%
YOLOv8l	9.664	89.04%	84.97%	89.72%	71.83%	46.85%
YOLOv8x	6.324	88.74%	84.26%	90.19%	71.52%	47.45%

Key finding: YOLOv8-nano achieved the best macro average on the test videos (81.21%) despite being the smallest model. The authors suggest this is because rip current detection relies on low-level visual patterns rather than complex high-level features — making a shallow network sufficient and a deeper one unnecessary.

Failure Cases

- DJI_0808.mp4 — near complete failure (<1% mAP) for YOLOv8-s, -m, and -x. Likely caused by unusual lighting, camera angle or severe glare on the water surface.

- no_rip_03.mp4 — strong false positive from YOLOv8-nano (only 12.49% accuracy). The video was taken at head-level elevation and the horizontal breaking wave pattern was mistakenly detected as a rip current.

- DJI_0423.mp4 — poor performance across most models, suggesting difficult conditions such as low contrast or heavily breaking waves obscuring the rip channel.

Bias and Generalisation Issues

- All footage was captured at a single beach in Romania. The model may not generalise to Australian beaches with different wave patterns, sand colour, water clarity or sun angle.

- Most training images are from Google Earth (satellite / aerial) — very different perspective from ground-level or drone footage at lower elevations.

- No night-time, stormy or heavily overcast conditions are included in the dataset.

6. Practical Deployment Considerations

- YOLOv8-nano runs at 73 FPS on an RTX 3090 — the only model capable of real-time 30 FPS video processing on high-end GPU hardware. Larger models drop well below 30 FPS.

- The paper explicitly states YOLOv8-nano is runnable on a portable device — making it suitable for mobile or edge deployment such as a beach-mounted camera system or drone with onboard compute.

- Input resolution: training at 640×640 pixels. Test videos at 1920×1080 (downsampled for inference). 4K video (3840×2160) was also included in the test set.

- No mobile-specific optimisation (quantisation, pruning) was performed — the nano model's speed comes from its small size, not hardware-specific optimisation.

- No real-time video pipeline was built — the model processes individual frames independently with no temporal smoothing or tracking between frames.

7. Key Takeaways for SurfWatch

3 Ideas to Adopt

- Use YOLOv8-nano as the starting baseline model. It outperforms all larger variants on real-world video test data, runs in real time, and is deployable on portable hardware. Start here before considering larger architectures.

- Use polygonal instance segmentation instead of bounding boxes. The paper clearly demonstrates that bounding boxes include too much irrelevant water and miss parts of the rip channel. Segmentation masks provide more precise, usable detections for safety alerts.

- Use mAP50 as the primary evaluation metric and report both macro and micro averages across test videos. This allows fair comparison between models and reveals per-video failure cases that aggregate metrics would hide.

2 Risks to Avoid

- Do not split by frames — must split by video or beach site. If the same rip current appears in both train and test frames, the model will appear far more accurate than it actually is. The paper avoids this by keeping training images and test videos completely separate.

- Do not claim real-world performance based on a single-location dataset. All footage is from one Romanian beach. SurfWatch must validate on Australian beach footage before any deployment claims — wave patterns, lighting, sand colour and camera setups will differ significantly.

Recommended Fit for SurfWatch MVP

- Video-based MVP — the paper's test set is video-based and drone-captured, which aligns with SurfWatch's likely deployment scenario of beach cameras or drone surveillance.

- Segmentation over bounding boxes or classification — instance segmentation provides the most precise and informative output for downstream safety alerts and lifeguard displays.

- Start with image-only training (as this paper does) then extend to video-aware temporal models once the baseline is validated — this is the same roadmap the authors suggest for future work.